MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT
TRANSISTORS, N-CHANNEL, SILICON
TYPES 2N6764, 2N6766, 2N6768, 2N6770,
JANTX, JANTXV, AND JANS

This amendment forms a part of MIL-S-19500/543C, dated 8 September 1987, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Upper left hand corner, implementation date, third line: Delete "8 September 1988" and substitute "8 May 1989".

PAGE 4

4.3, Screen 12, JANS column: Delete "test condition A and C (see 4.3.1)" and substitute "test condition A; (see 4.3.1) condition C shall precede condition A".

PAGE 5

4.3.1, second line: Delete "-0" and substitute "-5°C".

4.5.3, delete the main text only and substitute the following:

"The delta V_{SD} measurements shall be performed in accordance with method 3161 of MIL-STD-750. The delta V_{SD} conditions (I_H and V_H) and maximum V_{SD} limit shall be derived by each vendor from the thermal response curves (see figure 2). The chosen ΔV_{SD} measurement and conditions for each device in the qualification lot shall be submitted in the qualification report. The chosen ΔV_{SD} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots."

PAGE 6

TABLE I, Subgroup 2, Forward transconductance, Conditions column: Delete "Pulsed (see 4.5.1)" and substitute "I_D = Rated I_{D2} (see 1.3), pulsed (see 4.5.1)".

TABLE I, Subgroup 2, Forward transconductance: Move entire inspection to last place in subgroup 2.

TABLE I, Subgroup 2, Gate current, Conditions column: Delete "V_{GS} = ±20 V dc" and substitute "V_{GS} = +20 and -20 V dc".

PAGE 7

TABLE I, Subgroup 3, Gate current, Conditions column: Delete "V_{GS} = ±20 V dc" and substitute "V_{GS} = +20 V and -20 V dc".

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PAGES 7 AND 8

TABLE I, Subgroup 4, delete in its entirety and substitute the following:

Inspection	Method	MIL-STD-750 Conditions	LTPD 1/ JANS	LTPD 1/ JANTX, JANTXV	Symbol	Limits Min	Limits Max	Unit
Subgroup 4								
Switching time test	3472	I_D = Rated I_{D2} (see 1.3); V_{GS} = 10 V dc; Gate drive impedance = 2.35Ω						
Turn-on delay time					$t_{d(ON)}$			
2N6764		V_{DD} = 24 V dc				---	35	ns
2N6766		V_{DD} = 95 V dc				---	35	ns
2N6768		V_{DD} = 180 V dc				---	35	ns
2N6770		V_{DD} = 210 V dc				---	35	ns
Rise time					t_r			
2N6764		V_{DD} = 24 V dc				---	100	ns
2N6766		V_{DD} = 95 V dc				---	100	ns
2N6768		V_{DD} = 180 V dc				---	65	ns
2N6770		V_{DD} = 210 V dc				---	50	ns
Turn-off delay time					$t_{d(OFF)}$			
2N6764		V_{DD} = 24 V dc				---	125	ns
2N6766		V_{DD} = 95 V dc				---	125	ns
2N6768		V_{DD} = 180 V dc				---	150	ns
2N6770		V_{DD} = 210 V dc				---	150	ns
Fall time					t_f			
2N6764		V_{DD} = 24 V dc				---	100	ns
2N6766		V_{DD} = 95 V dc				---	100	ns
2N6768		V_{DD} = 180 V dc				---	75	ns
2N6770		V_{DD} = 210 V dc				---	70	ns

PAGE 8

TABLE I, Subgroup 5, delete entire subgroup and substitute the following:

Inspection	Method	MIL-STD-750 Conditions	LTPD 1/ JANS	LTPD 1/ JANTX, JANTXV	Symbol	Limits Min	Limits Max	Unit
Subgroup 5								
Safe operating area test	3474	See figure 3, V_{DS} = 80% of rated V_{DS} and $V_{DS} \leq$ 20 V maximum						
High voltage dc SOA		t_p = 1 s						
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7						

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PAGE 9

TABLE I, Subgroup 7, Test 1, Minimum off-state gate charge, 2N6768 and 2N6770, Min limits column: Delete "4.0" and substitute "3.0" (two places).

TABLE I, Subgroup 7, Test 2, On-state gate charge, Max limits column: Delete "119", "118", "120", "124" and substitute "125", "125", "125", "135".

PAGE 10

TABLE I, Subgroup 7, Test 6, Gate to drain charge, Max limits column: Delete "64", "65", "56", "61" and substitute "75", "75", "75", "80".

PAGE 11

TABLE IIa, Subgroup 4, Intermittent operation life, Conditions column: Delete ", see 4.3.1".

TABLE IIa, Subgroup 5, Accelerated steady-state operation life, Conditions column: Delete "T_J = 200°C" and substitute "(except) T_J = 200°C +20°C, -0°C".

PAGE 13

TABLE IIb, Subgroup 3, Intermittent operation life (LTPD), Conditions column: Delete "A" and substitute "The heating".

PAGE 14

TABLE III, Subgroup 2: Delete "Visual and mechanical evaluation" and method "2071".

TABLE III, Subgroup 6, Intermittent operation life (LTPD), Conditions column: Delete "A" and substitute "The heating".

PAGE 16

TABLE IV, Step 3, Gate current, Condition column: Delete "V_{GS} = ±20 V dc" and substitute "V_{GS} = +20 and -20 V dc".

TABLE IV, footnote 1/: Delete in its entirety.

PAGE 17

FIGURE 2: Delete ".050" and substitute "Single pulse". Also, NOTE 2: Delete "RθJC = 1.0" and substitute "$R_{\theta JC}$ = .83".

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Air Force - 11, 17, 80
DLA - ES
NASA - EG13

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1068)

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